

Exercise 08 - Streaming

Run kafka in a docker container

Start a Kafka broker using the official Apache Kafka image:

```
1 docker run -d -p 9092:9092 --name broker apache/kafka:latest
```

Test your setup

1. Connect to the Kafka container

```
1 docker exec --workdir /opt/kafka/bin/ -it broker sh
```

2. Create a topic

```
1 ./kafka-topics.sh --bootstrap-server localhost:9092 --create --topic  
  test-topic
```

3. Write events to the topic

```
1 ./kafka-console-producer.sh --bootstrap-server localhost:9092 --topic  
  test-topic
```

This command will wait for input at a > prompt. Enter `hello`, press Enter, then `world`, and press Enter again. Enter Ctrl+C to exit the console producer.

4. Read all the events from the topic

```
1 ./kafka-console-consumer.sh --bootstrap-server localhost:9092 --topic  
  test-topic --from-beginning
```

You should see the two strings you entered: hello and world.

Create a Kafka Producer

Create a kafka producer in python using kafka for python. The producer should use the random user api to store a fictional user in an kafka event ten times.

Use Python and the kafka-python library to create a Kafka producer. The producer should fetch data from the Random User API and send 10 fictional user records as Kafka events to the topic. The events should be stored in a new topic called `users-raw-data` this topic can be created manually as shown in the previous step.

Create a Kafka Consumer

Using the same Python library, create a Kafka consumer that reads events from the topic and prints the following string for each event:

```
1 Username: USERNAME, Email: EMAIL
```

Where USERNAME and EMAIL are extracted from the JSON content of each Kafka event.